

DEEP REINFORCEMENT LEARNING

HOMEWORK 3 | ACTOR-CRITIC METHODS

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1 – Deterministic Actor-Critic: From DPG to TD3

This problem studies deterministic policy gradients, DDPG, and TD3, emphasizing derivations, off-policy actor-critic learning, function-approximation error, overestimation bias, and algorithmic implications in continuous action spaces. The three coding tasks (DPG, DDPG, TD3) are addressed in the companion notebook.

1.1 (1 Point) Action-Value Gradient Derivation

Using the chain rule, derive $\nabla_{\theta} Q^{\mu_k}(s, \mu_{\theta}(s))$ and show it equals $\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu_k}(s, a)|_{a=\mu_{\theta}(s)}$.

Answer

Step 1: Identify the composition. Fix μ_k , so $Q^{\mu_k}(s, a)$ does not depend on θ . Write $a(\theta) = \mu_{\theta}(s) \in \mathbb{R}^n$ and $h(\theta) = Q^{\mu_k}(s, a(\theta))$; only the action argument carries θ .

Step 2: Apply the chain rule. For the scalar field $Q^{\mu_k}(s, \cdot) : \mathbb{R}^n \rightarrow \mathbb{R}$ composed with $a(\theta)$,

$$\nabla_{\theta} Q^{\mu_k}(s, \mu_{\theta}(s)) = (\nabla_{\theta} \mu_{\theta}(s)) (\nabla_a Q^{\mu_k}(s, a))|_{a=\mu_{\theta}(s)},$$

where $\nabla_{\theta} \mu_{\theta}(s) \in \mathbb{R}^{|\theta| \times n}$ is the policy Jacobian and $\nabla_a Q^{\mu_k} \in \mathbb{R}^n$ is the action-gradient of the critic.

Step 3: Conclusion.

$$\nabla_{\theta} Q^{\mu_k}(s, \mu_{\theta}(s)) = \nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu_k}(s, a)|_{a=\mu_{\theta}(s)}$$

No integration over actions is required: the gradient is a product of an actor quantity and a critic quantity.

1.2 (8 Points) Proof of the DPG Theorem

Prove $\nabla_{\theta} J(\mu_{\theta}) = \mathbb{E}_{s \sim \rho^{\mu}} [\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu}(s, a)|_{a=\mu_{\theta}(s)}]$ via the four steps (value-function form, $\nabla_{\theta} V = \nabla_{\theta} Q$, Bellman expansion, recursive unrolling).

Answer

Step 1: Express J through the value function. With start distribution ρ_0 and $V^{\mu_{\theta}}(s) = Q^{\mu_{\theta}}(s, \mu_{\theta}(s))$, both $\int_S \rho^{\mu_{\theta}}(s) r(s, \mu_{\theta}(s)) ds$ and $\int_S \rho_0(s_0) V^{\mu_{\theta}}(s_0) ds_0$ expand to the same discounted sum of rewards, using $\rho^{\mu_{\theta}}(s) = \sum_t \gamma^t \int \rho_0(s_0) p(s_0 \rightarrow s, t) ds_0$:

$$J(\mu_{\theta}) = \int_S \rho_0(s_0) V^{\mu_{\theta}}(s_0) ds_0$$

Step 2: From $V^{\mu_{\theta}}(s) = Q^{\mu_{\theta}}(s, \mu_{\theta}(s))$, differentiating gives $\nabla_{\theta} V^{\mu_{\theta}}(s) = \nabla_{\theta} Q^{\mu_{\theta}}(s, \mu_{\theta}(s))$.

Step 3: Bellman expansion. With $Q^{\mu_{\theta}}(s, a) = r(s, a) + \gamma \int p(s'|s, a) V^{\mu_{\theta}}(s') ds'$, the composite depends on θ through $a = \mu_{\theta}(s)$ in r , through a in p , and explicitly through $V^{\mu_{\theta}}$:

$$\begin{aligned} \nabla_{\theta} Q^{\mu_{\theta}}(s, \mu_{\theta}(s)) &= \nabla_{\theta} \mu_{\theta}(s) \nabla_a r|_{a=\mu_{\theta}(s)} + \gamma \nabla_{\theta} \mu_{\theta}(s) \int \nabla_a p(s'|s, a)|_{a=\mu_{\theta}(s)} V^{\mu_{\theta}}(s') ds' \\ &\quad + \gamma \int p(s'|s, \mu_{\theta}(s)) \nabla_{\theta} V^{\mu_{\theta}}(s') ds'. \end{aligned}$$

The first two terms combine into $\nabla_{\theta}\mu_{\theta}(s)\nabla_a Q^{\mu_{\theta}}(s, a)|_{a=\mu_{\theta}(s)}$, so

$$\nabla_{\theta}Q^{\mu_{\theta}}(s, \mu_{\theta}(s)) = \nabla_{\theta}\mu_{\theta}(s)\nabla_a Q^{\mu_{\theta}}(s, a)|_{a=\mu_{\theta}(s)} + \gamma \int p(s'|s, \mu_{\theta}(s))\nabla_{\theta}V^{\mu_{\theta}}(s')ds'$$

Remark. The prompt's $\nabla_a r$ is exact only when the transition kernel is action-independent; in general the leading term is the action-gradient of the whole $Q^{\mu_{\theta}}$, which is precisely what the final theorem contains.

Step 4: Unroll the recursion. With $b(s) \equiv \nabla_{\theta}\mu_{\theta}(s)\nabla_a Q^{\mu_{\theta}}(s, a)|_{a=\mu_{\theta}(s)}$ and $\nabla_{\theta}V^{\mu_{\theta}}(s) = b(s) + \gamma \int p(s \rightarrow s', 1, \mu_{\theta})\nabla_{\theta}V^{\mu_{\theta}}(s')ds'$, substituting repeatedly (Chapman–Kolmogorov) gives

$$\nabla_{\theta}V^{\mu_{\theta}}(s) = \int_S \sum_{t=0}^{\infty} \gamma^t p(s \rightarrow s', t, \mu_{\theta}) \nabla_{\theta}\mu_{\theta}(s')\nabla_a Q^{\mu_{\theta}}(s', a)|_{a=\mu_{\theta}(s')}ds'$$

Step 5: Assemble. Inserting into $\nabla_{\theta}J = \int \rho_0(s_0)\nabla_{\theta}V^{\mu_{\theta}}(s_0)ds_0$ and using $\rho^{\mu_{\theta}}(s') = \int \rho_0(s_0) \sum_t \gamma^t p(s_0 \rightarrow s', t)ds_0$,

$$\nabla_{\theta}J(\mu_{\theta}) = \mathbb{E}_{s \sim \rho^{\mu}} [\nabla_{\theta}\mu_{\theta}(s)\nabla_a Q^{\mu_{\theta}}(s, a)|_{a=\mu_{\theta}(s)}]. \quad \blacksquare$$

1.3 (2 Points) Deterministic On-Policy Actor-Critic

Derive the on-policy deterministic actor-critic updates with critic Q_w and exploration noise $a_t = \mu_{\theta}(s_t) + \mathcal{N}(0, \sigma^2)$.

Answer

Critic. With $\delta_t = r_t + \gamma Q_w(s_{t+1}, \mu_{\theta}(s_{t+1})) - Q_w(s_t, a_t)$,

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q_w(s_t, a_t).$$

Actor (DPG ascent).

$$\theta_{t+1} = \theta_t + \alpha_{\theta} \nabla_{\theta}\mu_{\theta}(s_t) \nabla_a Q_w(s, a)|_{a=\mu_{\theta}(s_t)}.$$

Exploration enters only data collection; the gradient uses the noise-free action $\mu_{\theta}(s_t)$.

1.4 (3 Points) Deterministic Off-Policy Actor-Critic

Derive the off-policy updates (target μ_{θ} , behavior β) and explain why no importance-sampling correction is needed in the actor.

Answer

Critic (off-policy, bootstrap toward the target action):

$$\delta_t = r_t + \gamma Q_w(s_{t+1}, \mu_{\theta}(s_{t+1})) - Q_w(s_t, a_t), \quad w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q_w(s_t, a_t).$$

Actor (no importance weight):

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu_\theta(s_t) \nabla_a Q_w(s, a)|_{a=\mu_\theta(s_t)}.$$

Why no IS. In the stochastic case the gradient is an expectation over $a \sim \pi_\theta$, so sampling from β requires the ratio $\rho_t = \pi_\theta/\beta$. In the deterministic case

$$\nabla_\theta J_\beta(\mu_\theta) \approx \mathbb{E}_{s \sim \rho^\beta} [\nabla_\theta \mu_\theta(s) \nabla_a Q^\mu(s, a)|_{a=\mu_\theta(s)}],$$

the inner action expectation has collapsed to the single point $a = \mu_\theta(s)$ (not the behaviour action $a_t \sim \beta$), so there is no action distribution to reweight. The behaviour policy only changes the state distribution $\rho^\mu \rightarrow \rho^\beta$, a mismatch the off-policy DPG simply accepts (as in OPDAG/DDPG).

1.5 (5 Points) Implementation 1: Deterministic Policy Gradient (DPG)

Implement a near-on-policy deterministic actor-critic baseline on the moving-target Reacher environment; plot learning curves and discuss stability/sample-efficiency relative to DDPG and TD3.

Answer

Setup. A deterministic tanh-squashed actor $a = b + s \tanh(f_\theta(o))$ and a Q -critic $Q_\phi(o, a)$, trained with the one-step DPG scheme: critic target $y_t = r_t + \gamma(1 - d_t)Q_\phi(o', \mu_\theta(o'))$ (termination masked only in the bootstrap), critic loss $L_Q = (Q_\phi(o, a) - y)^2$, and actor ascent on $-Q_\phi(o, \mu_\theta(o))$. Data are collected with additive exploration noise.

Two fixes found by tracing (not just running) the code.

- *Actor tanh saturation $\Rightarrow$ dead policy gradient.* Large pre-activations rail $\tanh$ to ± 1 , so ∇_a vanishes and training stays flat. Fixed with a small final-layer initialization ($\pm 3 \times 10^{-3}$) so initial actions start near 0.
- *Uncontrolled per-agent randomness.* `train_agent` reseeded only the environment, so comparisons measured initialization luck. Fixed by calling `set_global_seed()` before each agent.

Result. After 25k steps the near-on-policy DPG baseline fails to track: deterministic eval return ≈ -25.8 with mean fingertip–target distance ≈ 15 cm (Table 1, Figure 1). This matches the theory: with no replay buffer the updates use *correlated* sequential data, the bootstrap target $r + \gamma Q_\phi(o', \mu_\theta(o'))$ is a *moving* target (both ϕ and θ shift every step), and each transition is consumed once—so the method is both unstable and sample-inefficient relative to DDPG and TD3.

1.6 (3 Points) Proof of Compatible Function Approximation

Prove that under the compatibility conditions, the approximate DPG equals the true DPG.

Answer

Step 1 (Condition 1). The compatible critic satisfies $\nabla_a Q_w(s, a)|_{a=\mu_\theta(s)} = \nabla_\theta \mu_\theta(s)^\top w$ (e.g. $Q_w(s, a) = (a - \mu_\theta(s))^\top \nabla_\theta \mu_\theta(s)^\top w + V(s)$).

Step 2 (First-order optimality). w minimizes the mean-squared *gradient* error $\epsilon^2(w) = \mathbb{E}_{s \sim \rho^\mu} \|\nabla_a Q_w|_{a=\mu_\theta(s)} - \nabla_a Q^\mu|_{a=\mu_\theta(s)}\|^2$. Setting $\nabla_w \epsilon^2 = 0$ with $\nabla_w [\nabla_a Q_w|_{a=\mu_\theta(s)}] = \nabla_\theta \mu_\theta(s)^\top$,

$$\mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) (\nabla_a Q_w - \nabla_a Q^\mu)|_{a=\mu_\theta(s)}] = 0.$$

Remark. The unbiasedness condition is naturally on the action-gradients (this is what enters the DPG); the value-MSE wording should be read in this sense.

Step 3 (Combine).

$$\mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) \nabla_a Q_w|_{a=\mu_\theta(s)}] = \mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) \nabla_a Q^\mu|_{a=\mu_\theta(s)}]$$

Hence the compatible critic yields an unbiased policy gradient. ■

1.7 (3 Points) DDPG Component Analysis

For each component explain how it addresses the practical challenges, plus one limitation.

Answer

(a) Experience replay buffer. Reuses transitions (sample efficiency) and decorrelates the sequential stream (stability). *Limitation:* mixes data from old policies (stale off-policy targets), large memory, problematic under strong non-stationarity.

(b) Target networks with soft updates. Slowly tracked copies make the bootstrap target $r + \gamma Q_{w'}(s', \mu_{\theta'}(s'))$ change slowly, fixing the moving-target instability. *Limitation:* targets are biased (lag); too small τ slows learning, too large reintroduces instability.

(c) Batch normalization. Normalizes differing state scales / covariate shift, giving better-conditioned gradients. *Limitation:* train/eval running-statistics mismatch, batch-size sensitivity, drift under non-stationary RL data.

(d) Ornstein–Uhlenbeck noise. Temporally correlated exploration suited to inertial systems, addressing the lack of inherent exploration of deterministic policies. *Limitation:* extra hyperparameters; uncorrelated Gaussian noise is often as good; mis-tuning can under-explore or destabilize.

1.8 (5 Points) Implementation 2: Deep Deterministic Policy Gradient (DDPG)

Extend DPG to off-policy DDPG (replay buffer, target networks); compare against DPG.

Answer

Setup. DDPG extends DPG with a *replay buffer*, *target networks* $Q_{\bar{\phi}}, \mu_{\bar{\theta}}$ (Polyak-averaged), and temporally-correlated *Ornstein–Uhlenbeck* exploration. Critic target $y_t = r_t + \gamma(1 - d_t)Q_{\bar{\phi}}(o', \mu_{\bar{\theta}}(o'))$, critic loss $L_Q = \mathbb{E}_{(s,a,r,s',d) \sim D} [(Q_\phi - y)^2]$, actor objective $J(\theta) = \mathbb{E}_{s \sim D} [Q_\phi(s, \mu_\theta(s))]$; evaluation uses the noise-free policy.

Result. DDPG reaches eval return ≈ -1.99 and fingertip error ≈ 1.3 cm (Table 1), a jump of about +24 in return over DPG. This is the decisive step: the replay buffer reuses and *decorrelates* transitions (fixing DPG's sample-inefficiency and the correlation that violates the i.i.d. assumption of SGD), while the slowly-tracked target networks turn the moving bootstrap target into a near-stationary regression target (fixing the moving-target instability). Together they let the continuous actor-critic actually track the moving target.

1.9 (2 Points) Critic Gradient Error and Actor Bias

Prove the actor-gradient error is bounded by $M\epsilon$ and explain the sensitivity.

Answer

Let $\Delta(s) \equiv (\nabla_a Q_\phi - \nabla_a Q^\mu)|_{a=\mu_\theta(s)}$, $\|\Delta(s)\|_2 \leq \epsilon$. By Jensen and submultiplicativity,

$$\left\| \mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) \Delta(s)] \right\|_2 \leq \mathbb{E} \|\nabla_\theta \mu_\theta(s) \Delta(s)\|_2 \leq \mathbb{E} [\|\nabla_\theta \mu_\theta(s)\|_{\text{op}} \|\Delta(s)\|_2] \leq M\epsilon.$$

$$\left\| \mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) (\nabla_a Q_\phi - \nabla_a Q^\mu)|_{a=\mu_\theta(s)}] \right\|_2 \leq M\epsilon$$

Interpretation. Deterministic actor-critic differentiates the critic through the action: the update direction is the *slope* $\nabla_a Q_\phi$, not the value. A critic with small value error but a wrong action-gradient sends the policy in a wrong direction, amplified by the Jacobian norm M . Hence the need for TD3-style error control.

1.10 (2 Points) Overestimation and Policy Performance

Explain why overestimation harms the policy.

Answer

The actor maximizes Q_ϕ , so where Q_ϕ overestimates, the actor is driven toward those over-valued actions—it *exploits* the critic's errors. The realized return is lower than predicted, and since the inflated values seed the bootstrap targets $r + \gamma Q_\phi(s', \mu_\theta(s'))$, the overestimation propagates and accumulates. This self-reinforcing loop (actor chases the error peaks, peaks feed the targets) yields a suboptimal or diverging policy whose predicted value far exceeds its true performance.

1.11 (2 Points) Sources of Overestimation Bias

List three sources.

Answer

1. **Maximization over noisy estimates (Jensen gap):** $\mathbb{E}[\max_a \hat{Q}] \geq \max_a \mathbb{E}[\hat{Q}]$; the actor acts as the maximizer, biasing upward even for zero-mean noise.
2. **Function-approximation / generalization error:** the neural critic creates spurious action-space peaks that the maximization locks onto.
3. **Bootstrapping (error accumulation):** biased estimates at s' feed the target for s ; with discounting the bias compounds along the bootstrap chain rather than washing out.

1.12 (3 Points) Minimum of Two Q-Estimates

With $Q_{\phi_i} = Q^* + \epsilon_i$ (ϵ_i independent, zero-mean, symmetric), show $\mathbb{E}[\min_i Q_{\phi_i}] \leq Q^*$.

Answer

$\min_i Q_{\phi_i}(s, a) = Q^*(s, a) + \min(\epsilon_1, \epsilon_2)$. Using $\min(x, y) = \frac{1}{2}(x + y - |x - y|)$,

$$\mathbb{E}[\min(\epsilon_1, \epsilon_2)] = \frac{1}{2}(\mathbb{E}\epsilon_1 + \mathbb{E}\epsilon_2 - \mathbb{E}|\epsilon_1 - \epsilon_2|) = -\frac{1}{2}\mathbb{E}|\epsilon_1 - \epsilon_2| \leq 0.$$

$$\mathbb{E}[\min_{i=1,2} Q_{\phi_i}(s, a)] = Q^*(s, a) - \frac{1}{2}\mathbb{E}|\epsilon_1 - \epsilon_2| \leq Q^*(s, a)$$

with equality iff $\epsilon_1 = \epsilon_2$ a.s. (Only zero-mean is needed for the cancellation.) ■

1.13 (2 Points) Underestimation vs. Overestimation

Why is slight underestimation less harmful?

Answer

Overestimation lies on the side the maximizing actor climbs toward: an over-valued action is selected, its inflated value enters the bootstrap target, and the error is amplified (positive feedback, Q8). Underestimation points the other way: under-valued actions are simply *avoided*, so the pessimistic error is not exploited or amplified and creates no spurious attractors. A roughly uniform downward bias also preserves action *rankings*, so the greedy choice is often still correct—merely conservative. Thus the deliberate pessimism of clipped double-Q trades a benign bias for the removal of a dangerous one.

1.14 (2 Points) Delayed Policy Updates

Explain the harm of too-frequent actor updates; interpret delay as two-time-scale.

Answer

If the actor updates as fast as the critic, it chases a not-yet-converged, high-variance critic ($\nabla_a Q_\phi$ frequently wrong), and each actor change shifts the critic's target, causing coupled oscillation. Two-time-scale stochastic approximation is well-behaved when one component is fast and the other slow: the critic should converge to Q^{μ_θ} for a roughly fixed θ (fast), the actor should see a nearly converged critic (slow). Updating the actor once every d critic steps ($d = 2$) approximately realizes this separation, reducing variance and breaking the feedback—a practical surrogate for $\alpha_\theta/\alpha_w \rightarrow 0$.

1.15 (2 Points) Target Policy Smoothing Intuition

Why does clipped noise on the target action help?

Answer

A deterministic target action evaluates the critic at a single point, letting the actor exploit a sharp, narrow, spurious peak of Q_ϕ (overfitting in action space). Adding clipped noise and using $Q_{\phi'}(s', \mu_{\theta'}(s') + \epsilon)$ replaces the point by a local average, encoding the prior that similar actions have similar values. It smooths Q in action space, lowers target variance, and flattens narrow erroneous spikes so the actor cannot latch onto them—an Expected-SARSA-like regularized backup.

1.16 (3 Points) Target Smoothing and Regularization

With $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, derive a second-order approximation of $\mathbb{E}_\epsilon[Q(s', \mu(s') + \epsilon)]$ and relate it to regularization.

Answer

Step 1 (Taylor).

$$Q(s', \mu(s') + \epsilon) \approx Q(s', \mu(s')) + \nabla_a Q(s', \mu(s'))^\top \epsilon + \frac{1}{2} \epsilon^\top \nabla_a^2 Q(s', \mu(s')) \epsilon.$$

Step 2 (Expectation). Using $\mathbb{E}[\epsilon] = 0$, $\mathbb{E}[\epsilon^\top H \epsilon] = \sigma^2 \text{tr}(H)$,

$$\mathbb{E}_\epsilon[Q(s', \mu(s') + \epsilon)] \approx Q(s', \mu(s')) + \frac{\sigma^2}{2} \text{tr}(\nabla_a^2 Q(s', \mu(s')))$$

Step 3 (Interpretation). The smoothed target adds a curvature penalty $\frac{\sigma^2}{2} \text{tr}(\nabla_a^2 Q)$ (a Laplacian in action space). At a sharp maximum the Hessian is negative-definite, so the target is pulled *down* exactly where the critic spikes—a Tikhonov/Laplacian smoothing that discourages high action-space curvature and counteracts overestimated peaks.

1.17 (10 Points) Implementation 3: Twin Delayed DDPG (TD3)

Extend DDPG to TD3 (clipped double-Q, delayed updates, target smoothing); compare with DDPG and DPG, with at least one ablation.

Answer

Setup. TD3 adds the three mechanisms to DDPG: *clipped double-Q* ($y = r + \gamma(1 - d) \min_{i=1,2} Q_{\bar{\phi}_i}(o', a')$), *target-policy smoothing* ($a' = \text{clip}(\mu_{\bar{\theta}}(o') + \epsilon, a_{\min}, a_{\max})$ with $\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c)$), and *delayed actor/target updates* (every $d = 2$ critic steps), with actor loss $-\mathbb{E}[Q_{\phi_1}(o, \mu_\theta(o))]$.

Results. Full TD3 attains the best deterministic evaluation: return ≈ -1.72 and fingertip error ≈ 1.1 cm. The eval summary is in Table 1; learning curves for all agents and the three single-mechanism ablations are in Figure 1.

Table 1: Deterministic evaluation after 25k steps/agent (seeded).

Agent	Eval return	Fingertip error
DPG	-25.8	≈ 15 cm (never tracks)
DDPG	-1.99	≈ 1.3 cm
TD3 (full)	-1.72	≈ 1.1 cm (best)

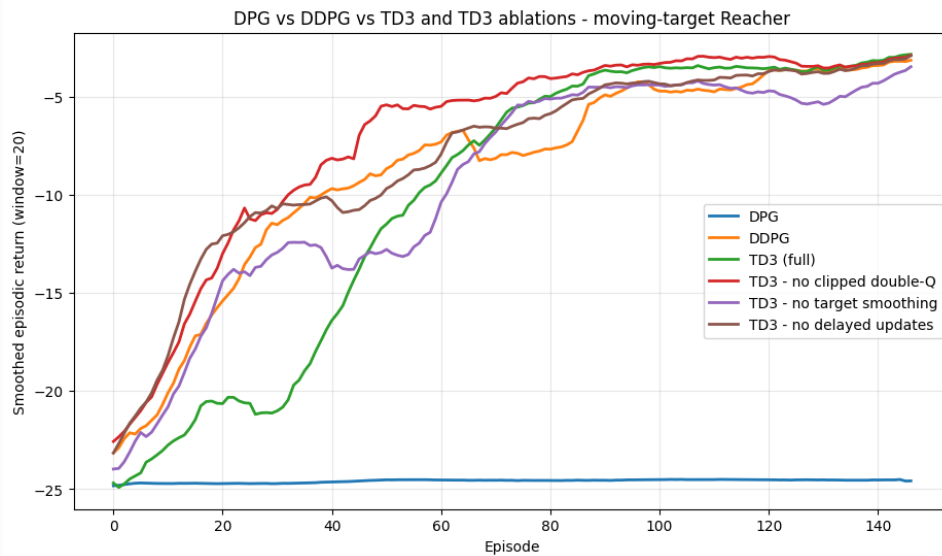


Figure 1: Learning curves for DPG, DDPG, full TD3, and the three TD3 ablations (no clipped double-Q, no target smoothing, no delayed updates); 25k steps/agent, seeded.

Ablations (train return, last 20 episodes): full TD3 -2.82 ; no target smoothing -3.45 ; no delayed updates -2.89 ; no clipped double-Q -2.86 .

Discussion. The dominant effect is *not* a TD3 component but the DDPG ingredients (replay buffer + target networks): the DPG $\rightarrow$ DDPG jump of $\approx +24$ return dwarfs the differences among the TD3 variants, which all track to roughly 0.6–1.1 cm. On this easy, densely-rewarded task TD3’s conservatism is barely needed: by deterministic eval, *removing* clipped double-Q actually tracked best (≈ 5.7 mm vs ≈ 11 mm for full TD3), because with little overestimation to correct the min-of-two-critics is merely pessimistic and slightly slows the critic. These mechanisms are insurance that pays off on harder, sparser, or higher-dimensional tasks where overestimation and the actor’s exploitation of critic errors (Questions 1.8–1.14) become severe. (Single seed; the gaps among the TD3 variants are small and within noise.)

2 – Soft Actor-Critic

Maximum-entropy RL augments return with an entropy regularizer, $J(\pi) = \sum_t \mathbb{E}_{(s_t, a_t) \sim \rho_\pi} [r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot|s_t))]$, with soft value $V^\pi(s_t) = \mathbb{E}_{a \sim \pi} [Q^\pi(s_t, a) - \alpha \log \pi(a|s_t)]$ and soft Bellman equation $Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s'} [V^\pi(s')]$.

2.1 (3 Points) Maximum Entropy Framework

Explain why entropy improves exploration and robustness.

Answer

(1) No premature collapse. The term $\alpha \mathcal{H}$ penalizes near-deterministic policies, so the optimizer keeps the policy stochastic until the reward advantage of committing genuinely outweighs the lost entropy—crucial early when value estimates are unreliable.

(2) Multiple near-optimal actions. When several actions have similar Q , the optimum is the Boltzmann distribution $\pi^* \propto \exp(Q/\alpha)$, spreading mass over all of them; this captures multi-modality, preserves options, and broadens exploration.

(3) Robustness. A sharply deterministic policy is brittle—a small Q error flips the argmax. A high-entropy policy hedges across actions so behavior/value vary smoothly with critic error; formally it coincides with a policy robust to a class of adversarial reward perturbations.

2.2 (5 Points) Soft Value Function and the Variational Formula

Show $V_Q(s) = \alpha \log \int_{\mathcal{A}} \exp(\frac{1}{\alpha} Q(s, a)) da$, derive the Boltzmann maximizer, and explain why $V_Q \neq V^\pi$ for arbitrary π .

Answer

Step 1 (Lagrangian). Maximize $\int \pi(a)(Q(s, a) - \alpha \log \pi(a)) da$ s.t. $\int \pi = 1$:

$$\mathcal{L}[\pi] = \int \pi(a)(Q(s, a) - \alpha \log \pi(a)) da + \lambda(1 - \int \pi(a) da).$$

Step 2 (Stationarity). $Q(s, a) - \alpha \log \pi(a) - \alpha - \lambda = 0 \Rightarrow \pi(a) \propto \exp(\frac{1}{\alpha} Q(s, a))$; the objective is concave, so this is the global max.

Step 3 (Boltzmann policy).

$$\pi_Q^*(a|s) = \frac{\exp(\frac{1}{\alpha} Q(s, a))}{\int_{\mathcal{A}} \exp(\frac{1}{\alpha} Q(s, \bar{a})) d\bar{a}}$$

Step 4 (Evaluate). With $Z(s) = \int \exp(\frac{1}{\alpha} Q) da$, $\log \pi_Q^* = \frac{1}{\alpha} Q - \log Z$, so $Q - \alpha \log \pi_Q^* = \alpha \log Z(s)$ (independent of a):

$$V_Q(s) = \alpha \log \int_{\mathcal{A}} \exp(\frac{1}{\alpha} Q(s, a)) da$$

Step 5. $V_Q(s) = \sup_{\pi} \mathbb{E}_{\pi}[Q - \alpha \log \pi]$ is the value of the *entropy-optimal* policy (a log-sum-exp), whereas $V^{\pi}(s) = \mathbb{E}_{a \sim \pi}[Q - \alpha \log \pi]$ for a fixed π . By the supremum, $V^{\pi} \leq V_Q$ with equality iff $\pi = \pi_Q^*$; otherwise strict.

2.3 (2 Points) Analysis on α

Analyze $\alpha \rightarrow 0$ and $\alpha \rightarrow \infty$.

Answer

$\alpha \rightarrow 0^+$ (**greedy**). $\alpha \log \int \exp(\frac{1}{\alpha} Q) da \rightarrow \max_a Q(s, a)$ and $\pi_Q^* \rightarrow$ deterministic greedy on $\arg \max_a Q$; standard RL recovered.

$\alpha \rightarrow \infty$ (**uniform**). Expanding $\exp(\frac{1}{\alpha} Q) \approx 1 + \frac{1}{\alpha} Q$,

$$\alpha \log \int_{\mathcal{A}} \exp(\frac{1}{\alpha} Q) da = \alpha \log \text{Vol}(\mathcal{A}) + \frac{1}{\text{Vol}(\mathcal{A})} \int_{\mathcal{A}} Q da + O(1/\alpha),$$

dominated by the entropy term $\alpha \log \text{Vol}(\mathcal{A})$ (or $\alpha \log |\mathcal{A}|$ if finite), with $\pi_Q^* \rightarrow \text{uniform}$. So α interpolates between exploitation and pure exploration.

2.4 (3 Points) Soft Policy Evaluation Lemma

Prove $Q_{k+1} = \mathcal{T}^\pi Q_k \rightarrow Q^\pi$.

Answer

Write $\mathcal{T}^\pi Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s', a'} [Q(s', a') - \alpha \log \pi(a'|s')]$. For any Q_1, Q_2 the reward and entropy terms cancel:

$$|\mathcal{T}^\pi Q_1(s, a) - \mathcal{T}^\pi Q_2(s, a)| = \gamma |\mathbb{E}_{s', a'} [Q_1 - Q_2]| \leq \gamma \|Q_1 - Q_2\|_\infty,$$

so $\mathcal{T}^\pi$ is a γ -contraction in $\|\cdot\|_\infty$. With $\gamma < 1$ and bounded augmented reward ($|\mathcal{A}| < \infty$), Banach's theorem gives a unique fixed point Q^π and $\|Q_k - Q^\pi\|_\infty \leq \gamma^k \|Q_0 - Q^\pi\|_\infty \rightarrow 0$. ■

2.5 (4 Points) Soft Bellman Residual Bound

If $\|\mathcal{T}^\pi Q - Q\|_\infty \leq \epsilon$, prove $\|Q - Q^\pi\|_\infty \leq \epsilon/(1 - \gamma)$.

Answer

Using $Q^\pi = \mathcal{T}^\pi Q^\pi$ and the γ -contraction,

$$\|Q - Q^\pi\|_\infty \leq \|Q - \mathcal{T}^\pi Q\|_\infty + \|\mathcal{T}^\pi Q - \mathcal{T}^\pi Q^\pi\|_\infty \leq \epsilon + \gamma \|Q - Q^\pi\|_\infty,$$

hence $(1 - \gamma)\|Q - Q^\pi\|_\infty \leq \epsilon$ and

$$\|Q - Q^\pi\|_\infty \leq \frac{\epsilon}{1 - \gamma} \quad \blacksquare$$

2.6 (3 Points) Soft Policy Improvement Lemma

Prove $Q^{\pi_{new}} \geq Q^{\pi_{old}}$ for the KL-projected π_{new} .

Answer

Step 1. With $J_{\pi_{old}}(\pi') = \mathbb{E}_{a \sim \pi'} [\alpha \log \pi'(a|s_t) - Q^{\pi_{old}}(s_t, a)]$ and $\alpha D_{\text{KL}}(\pi' \| e^{Q^{\pi_{old}}/\alpha} / Z) = J_{\pi_{old}}(\pi') + \alpha \log Z^{\pi_{old}}(s_t)$, the minimizer π_{new} obeys $J_{\pi_{old}}(\pi_{new}) \leq J_{\pi_{old}}(\pi_{old}) = -V^{\pi_{old}}(s_t)$, i.e.

$$\mathbb{E}_{a \sim \pi_{new}} [Q^{\pi_{old}}(s_t, a) - \alpha \log \pi_{new}(a|s_t)] \geq V^{\pi_{old}}(s_t). \quad (*)$$

Step 2 (Telescoping). Apply the soft Bellman equation and (*) repeatedly:

$$\begin{aligned} Q^{\pi_{old}}(s_t, a_t) &= r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}} [V^{\pi_{old}}(s_{t+1})] \\ &\leq r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}} [\mathbb{E}_{a_{t+1} \sim \pi_{new}} [Q^{\pi_{old}}(s_{t+1}, a_{t+1}) - \alpha \log \pi_{new}(a_{t+1}|s_{t+1})]] \\ &\leq \dots \leq Q^{\pi_{new}}(s_t, a_t), \end{aligned}$$

converging by soft policy evaluation. Hence $Q^{\pi_{new}}(s_t, a_t) \geq Q^{\pi_{old}}(s_t, a_t) \forall (s_t, a_t)$. ■

2.7 (3 Points) Soft Policy Iteration Theorem

Prove convergence to the optimal π^* .

Answer

Step 1. By Q6, $Q^{\pi_{i+1}} \geq Q^{\pi_i}$ pointwise, so $\{Q^{\pi_i}\}$ is monotone non-decreasing; with bounded rewards, $|\mathcal{A}| < \infty$, $\gamma < 1$ it is bounded above, hence converges to some Q^{π^*} .

Step 2. At convergence the improvement step achieves the minimizing Boltzmann distribution, so (*) holds for π^* against every $\pi \in \Pi$; repeating the telescoping argument with $\pi_{old} = \pi^*$, $\pi_{new} = \pi$ gives $Q^{\pi^*}(s, a) \geq Q^\pi(s, a)$ for all $\pi \in \Pi$, (s, a) . ■

2.8 (3 Points) SAC Actor Objective as KL Minimization

Show $J_\pi(\theta)$ equals (up to a state-dependent constant) the expected KL to the Boltzmann policy; explain the connection.

Answer

With $Z_\phi(s) = \int \exp(\frac{1}{\alpha} Q_\phi(s, a)) da$,

$$\alpha D_{\text{KL}}\left(\pi_\theta(\cdot|s) \parallel \frac{e^{Q_\phi/\alpha}}{Z_\phi}\right) = \mathbb{E}_{a \sim \pi_\theta} [\alpha \log \pi_\theta(a|s) - Q_\phi(s, a)] + \alpha \log Z_\phi(s).$$

Taking $\mathbb{E}_{s \sim D}$, $J_\pi(\theta) = \alpha \mathbb{E}_{s \sim D} [D_{\text{KL}}(\cdot \parallel \cdot)] - \alpha \mathbb{E}_{s \sim D} [\log Z_\phi(s)]$. Since $\log Z_\phi(s)$ is θ -independent and $\alpha > 0$,

$$\arg \min_{\theta} J_\pi(\theta) = \arg \min_{\theta} \mathbb{E}_{s \sim D} \left[D_{\text{KL}}(\pi_\theta(\cdot|s) \parallel \frac{1}{Z_\phi(s)} e^{Q_\phi(s, \cdot)/\alpha}) \right]$$

Connection. This is the soft policy improvement step (Q6) with $Q^{\pi_{old}}$ replaced by the critic Q_ϕ and the minimization restricted to $\{\pi_\theta\}$: SAC projects the policy (information/KL projection) onto the critic's Boltzmann distribution.

2.9 (2 Points) REINFORCE Variance

Explain why REINFORCE has high variance.

Answer

The score-function estimator $\mathbb{E}[\nabla_\theta \log \pi_\theta(a|s) R]$ uses the random return R only as a scalar weight, ignoring derivative information; its variance scales like $\text{Var}(R) \cdot \mathbb{E} \|\nabla \log \pi\|^2$. Pathwise (reparameterized) gradients differentiate through Q and the sample, $\nabla_a Q \nabla_\theta f_\theta$, exploiting analytic slope and giving lower variance. Long horizons worsen REINFORCE: $R = \sum_t \gamma^t r_t$ and the score $\sum_t \nabla \log \pi$ each sum many random terms, so their product's variance grows with T (every action is weighted by the entire noisy return, including rewards it did not influence); pathwise gradients are far less horizon-sensitive.

2.10 (5 Points) Reparameterized Gradient

Derive the full reparameterized gradient of $J_\pi(\theta)$ with $a_t = f_\theta(\epsilon_t; s_t)$.

Answer

Step 1. Since ϵ 's distribution is θ -independent, $\nabla_{\theta} J_{\pi} = \mathbb{E}_{s, \epsilon} [\nabla_{\theta} (\alpha \log \pi_{\theta}(a_t | s_t) - Q_{\phi}(s_t, a_t))]$ with $a_t = f_{\theta}(\epsilon_t; s_t)$.

Step 2 (Chain rule). The log-term has explicit and pathwise θ -dependence; Q_{ϕ} only pathwise:

$$\nabla_{\theta} [\alpha \log \pi_{\theta}(a_t | s_t)] = \alpha \nabla_{\theta} \log \pi_{\theta}(a | s_t)|_{a=a_t} + \alpha \nabla_a \log \pi_{\theta}(a | s_t)|_{a=a_t} \nabla_{\theta} f_{\theta},$$

$$\nabla_{\theta} [-Q_{\phi}(s_t, a_t)] = -\nabla_a Q_{\phi}(s_t, a)|_{a=a_t} \nabla_{\theta} f_{\theta}.$$

Step 3 (Full gradient).

$$\nabla_{\theta} J_{\pi}(\theta) = \mathbb{E}_{s, \epsilon} [\alpha \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) + (\alpha \nabla_a \log \pi_{\theta}(a_t | s_t) - \nabla_a Q_{\phi}(s_t, a_t)) \nabla_{\theta} f_{\theta}(\epsilon_t; s_t)]$$

Reconciliation. Reading $\nabla_{\theta} \alpha \log \pi_{\theta}(a_t | s_t)$ in Eq. (6) as the *total* derivative of the log-term (explicit + pathwise) makes (6) identical to the boxed result, whose second term is the pathwise gradient through the critic (matching Haarnoja et al., Eq. 13).

2.11 (3 Points) Squashed Gaussian Policy and Change of Variables

Derive the tanh log-probability correction.

Answer

With $u \sim \mathcal{N}(\mu_{\theta}(s), \sigma_{\theta}(s)^2)$ and $a = \tanh(u)$, the change of variables gives $\pi(a|s) = p_u(u) |\det(\partial a / \partial u)|^{-1}$. Since $a_j = \tanh(u_j)$ is elementwise, $\partial a / \partial u$ is diagonal with entries $1 - \tanh^2(u_j)$, so $|\det| = \prod_j (1 - \tanh^2(u_j))$. Hence

$$\log \pi_{\theta}(a|s) = \log \mathcal{N}(u; \mu_{\theta}(s), \sigma_{\theta}(s)^2) - \sum_{j=1}^d \log(1 - \tanh^2(u_j))$$

accounting for the volume contraction of tanh near saturation. (Numerically, $\log(1 - \tanh^2 u_j) = 2(\log 2 - u_j - \text{softplus}(-2u_j))$.)

2.12 (3 Points) Automatic Temperature Tuning

Derive $J(\alpha)$, $\nabla_{\alpha} J$, and the update behaviour.

Answer

Step 1 (Dual). For the constraint $\mathbb{E}[-\log \pi_{\theta}] \geq \bar{\mathcal{H}}$ with multiplier $\alpha \geq 0$, minimizing

$$J(\alpha) = \mathbb{E}_{s \sim D, a \sim \pi_{\theta}} [-\alpha (\log \pi_{\theta}(a|s) + \bar{\mathcal{H}})], \quad \alpha > 0$$

Step 2 (Gradient). $\nabla_{\alpha} J(\alpha) = \mathbb{E}[-(\log \pi_{\theta} + \bar{\mathcal{H}})] = \mathcal{H}_{\text{curr}} - \bar{\mathcal{H}}$ (with $\mathcal{H}_{\text{curr}} = \mathbb{E}[-\log \pi_{\theta}]$).

Step 3 (Behaviour). $\alpha \leftarrow \alpha - \lambda(\mathcal{H}_{\text{curr}} - \bar{\mathcal{H}})$. If too deterministic ($\mathcal{H}_{\text{curr}} < \bar{\mathcal{H}}$), $\nabla_{\alpha} J < 0 \Rightarrow \alpha$ increases (more exploration); if too stochastic ($\mathcal{H}_{\text{curr}} > \bar{\mathcal{H}}$), α decreases. A self-correcting controller driving entropy to $\bar{\mathcal{H}}$.

3 — Asynchronous Advantage Actor-Critic (A3C)

We study A2C/A3C: variance reduction, multi-step bootstrapping, entropy regularization, stale gradients, and parallel optimization.

3.1 (7 Points) Baselines, Bias, and the Optimal Baseline

(a) Show $\mathbb{E}_{a \sim \pi_\theta} [g_\theta(s, a)b(s)] = 0$. (b) Derive the optimal scalar $b^*(s)$. (c) Is $b^* = V^{\pi_\theta}$?

Answer

(a) With $g_\theta(s, a) = \nabla_\theta \log \pi_\theta(a|s)$,

$$\mathbb{E}_{a \sim \pi_\theta} [g_\theta(s, a)b(s)] = b(s) \nabla_\theta \int \pi_\theta(a|s) da = b(s) \nabla_\theta 1 = 0.$$

(b) Minimizing $\mathbb{E}_{a \sim \pi_\theta} [\|g_\theta\|^2 (Q^{\pi_\theta} - b)^2]$: $\frac{d}{db} = -2\mathbb{E}[\|g_\theta\|^2 (Q^{\pi_\theta} - b)] = 0$, so

$$b^*(s) = \frac{\mathbb{E}_{a \sim \pi_\theta} [\|g_\theta(s, a)\|_2^2 Q^{\pi_\theta}(s, a)]}{\mathbb{E}_{a \sim \pi_\theta} [\|g_\theta(s, a)\|_2^2]}$$

(c) Not in general: b^* is a $\|g_\theta\|^2$ -weighted average of Q , while $V^{\pi_\theta} = \mathbb{E}_a[Q]$ is unweighted. They coincide iff $\|g_\theta\|^2$ and Q^{π_θ} are uncorrelated under $\pi_\theta(\cdot|s)$. **Sufficient condition:** $\|\nabla_\theta \log \pi_\theta(a|s)\|^2$ is constant in a , giving $b^*(s) = V^{\pi_\theta}(s)$.

3.2 (3 Points) Multi-Step Advantage as a Sum of TD Errors

Show $\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}$ and discuss n .

Answer

With $\delta_{t+k} = r_{t+k} + \gamma V_\phi(s_{t+k+1}) - V_\phi(s_{t+k})$,

$$\sum_{k=0}^{n-1} \gamma^k \delta_{t+k} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \underbrace{[\gamma^n V_\phi(s_{t+n}) - V_\phi(s_t)]}_{\text{telescoping}} = R_t^{(n)} - V_\phi(s_t) = \hat{A}_t^{(n)}.$$

$$\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}$$

Small n : low variance, high (bootstrap) bias. Large n : low bias, high variance. n is the bias-variance knob.

3.3 (5 Points) Critic-Induced Bias in Multi-Step Actor-Critic

With $V_\phi = V^\pi + e$, derive the bias and answer the four sub-questions.

Answer

Derivation. Using $Q^\pi(s, a) = \mathbb{E}[\sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V^\pi(s_{t+n}) | s, a]$ and $V_\phi = V^\pi + e$,

$$\mathbb{E}[\hat{A}_t^{(n)} | s, a] = A^\pi(s, a) + \gamma^n \mathbb{E}[e(s_{t+n}) | s, a] - e(s).$$

With $\mathbb{E}[e(s_{t+n})|s, a] = (P(\cdot|s, a)P_\pi^{n-1}e)(s)$,

$$\text{Bias} = \gamma^n (P(\cdot|s, a)P_\pi^{n-1}e)(s) - e(s)$$

- (1) $n = 1$: $\text{Bias} = \gamma \sum_{s'} P(s'|s, a) e(s') - e(s)$. (2) $n \rightarrow \infty$: if $\gamma^n e \rightarrow 0$, $\text{Bias} \rightarrow -e(s)$.
 (3) $-e(s)$ depends only on the state, so it is a baseline: $\mathbb{E}_a[g_\theta(s, a)(-e(s))] = 0$ (Q1a)—no gradient bias. (4) $\gamma^n \mathbb{E}[e(s_{t+n})|s, a]$ depends on the *action* via $P(\cdot|s, a)$, so it correlates with $g_\theta(s, a)$ and biases the actor gradient.

3.4 (3 Points) A2C Update Equations and Semi-Gradient Critic Learning

Derive the critic and actor updates and explain "semi-gradient".

Answer

- (a) Treating $\hat{R}_t$ constant, $\nabla_\phi L_{\text{critic}}(\phi) = \mathbb{E}[(V_\phi(s_t) - \hat{R}_t) \nabla_\phi V_\phi(s_t)] = -\mathbb{E}[\hat{A}_t^{(n)} \nabla_\phi V_\phi(s_t)]$.
 (b) Treating $\hat{A}_t^{(n)}$ constant, $\nabla_\theta J_{\text{actor}}(\theta) = \mathbb{E}[\nabla_\theta \log \pi_\theta(a_t|s_t) \hat{A}_t^{(n)}]$.
 (c) The bootstrap target $\hat{R}_t$ contains $\gamma^n V_\phi(s_{t+n})$ and depends on ϕ , but is treated as constant—a *semi-gradient*. A full gradient would add $+\gamma^n \mathbb{E}[(V_\phi(s_t) - \hat{R}_t) \nabla_\phi V_\phi(s_{t+n})]$.

3.5 (2 Points) Entropy as KL Regularization

(a) Entropy = KL to uniform; (b) derive $\nabla_\theta H$; (c) importance in A3C.

Answer

- (a) $D_{\text{KL}}(\pi_\theta \| \text{Unif}) = -H(\pi_\theta) + \log |\mathcal{A}|$, so maximizing $H \Leftrightarrow$ minimizing the KL.
 (b) Using $\nabla_\theta (\pi \log \pi) = \nabla_\theta \pi (\log \pi + 1)$,

$$\nabla_\theta H(\pi_\theta(\cdot|s)) = - \sum_a \nabla_\theta \pi_\theta(a|s) (\log \pi_\theta(a|s) + 1)$$

- (c) A3C has no replay buffer; each worker learns from highly correlated on-policy data, risking premature collapse. The entropy bonus maintains exploration and prevents early collapse—the role replay plays off-policy.

3.6 (3 Points) Stale Gradients in Asynchronous Actor-Critic

Prove $\|G(\theta_t) - G(\theta_{t-\tau})\|_2 \leq L\alpha C\tau$ and discuss.

Answer

By L -Lipschitzness and the telescoping displacement,

$$\|G(\theta_t) - G(\theta_{t-\tau})\|_2 \leq L\|\theta_t - \theta_{t-\tau}\|_2 \leq L \sum_{k=t-\tau}^{t-1} \|\theta_{k+1} - \theta_k\|_2 \leq L\tau\alpha C.$$

$$\|G(\theta_t) - G(\theta_{t-\tau})\|_2 \leq L\alpha C\tau$$

The error is linear in α , τ , C . Smaller α shrinks it but slows learning; more parallel workers increase the effective delay τ , enlarging the error, so worker count and α must be balanced.

3.7 (5 Points) Decorrelation Through Parallel Workers

Compute $\text{Var}(\bar{g})$, analyze $\rho \in \{0, 1\}$, and explain decorrelation.

Answer

(a)

$$\text{Var}(\bar{g}) = \frac{1}{N^2} [N\Sigma + N(N-1)\rho\Sigma] = \boxed{\frac{1 + (N-1)\rho}{N}\Sigma}$$

(b) $\rho = 0$: $\text{Var}(\bar{g}) = \Sigma/N$ (full $1/N$ reduction). $\rho = 1$: $\text{Var}(\bar{g}) = \Sigma$ (no reduction).

(c) A3C runs each worker on its own environment with independent seeds, keeping ρ small, so $\text{Var}(\bar{g}) \approx \Sigma/N$. Parallel actors thus average out the noise of correlated single-stream data—playing the decorrelation role of experience replay, without storing data or introducing off-policy bias.